

Scalability & Future Plan

Date	29 September 2026
Team ID	6ab4cf016603573ccfb54f03
Project Name	AI-WeatherWise
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Depends on external weather and AI APIs	Service availability and response depend on external APIs	High
2	Local/server-based deployment	Limited ability to handle a large number of simultaneous users	Medium
3	Limited advanced weather analytics	Current system mainly provides current weather, summaries and recommendations	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for small-scale usage	Deploy using cloud infrastructure and add load balancing for larger user traffic
2	Data Storage	MongoDB database	Use scalable cloud MongoDB with indexing and optimized queries
3	Performance	Node.js/Express backend running as a local/server application	Use caching, optimized API calls and horizontal scaling
4	Security	JWT-based authentication	Add stronger access controls, HTTPS, rate limiting and improved security monitoring

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Weather forecasting and historical weather analysis	2–3 months	Provide more useful weather insights
Phase 3	Personalized notifications and alerts	3–4 months	Notify users about important weather change
Phase 4	Mobile application and advanced AI features	4–6 months	Expand accessibility and improve personalization